

PROJECT REVIEW – 2

Explainable Vision-Language Diagnostic Assistant for Chest Radiographs with Offline Rural Deployment

NLP Specialization Capstone Project

Cluster – 3 | Section – 7

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1. PROBLEM STATEMENT

- **Acute Radiologist Shortage:** Primary health centers in rural regions lack specialist radiologists to read X-rays and detect conditions like pneumonia or tuberculosis.
- **Diagnostic Delays & Travel Burden:** Patients may travel vast distances to urban hospitals, delaying time-critical care and increasing financial burden.
- **Black-Box AI Mistrust:** Traditional AI systems provide class outputs without explanations, making local nurses reluctant to rely on machine predictions.

The project addresses the need for an explainable diagnostic-support workflow that combines chest X-ray classification, visual evidence, text-based reporting and priority guidance for rural healthcare use.

2. OBJECTIVES

Objective	Description
1. Multimodal Diagnosis	Develop a vision-language assistant to generate plain-language diagnostic descriptions such as "early signs of pneumonia" alongside class confidence scores.
2. Visual Explanation	Incorporate Grad-CAM heatmaps and attention overlays that identify the anatomical regions contributing to the model output.
3. Automated Triage & Priority Guidance	Compute case-urgency flags, such as "Please refer to specialist within 24 hours", to help rural health workers prioritize critical cases.

3. LITERATURE SURVEY & RESEARCH GAP

3.1 Literature Survey – Existing Work

Existing Work	Contribution	Limitation
Standard CNN Classifiers – ResNet & DenseNet	Strong image-classification performance on chest X-rays such as ChestX-ray14.	Primarily classification-oriented and lacks integrated report generation and multimodal explanation.
Large Vision-Language Models	Can generate clinical-style reports by combining image and language information.	Heavy Transformer architectures can require GPU servers and stable cloud APIs, making offline edge deployment difficult.
Standalone Saliency / Heatmap Methods	Highlight image regions that contribute to model predictions.	Do not by themselves explain clinical severity or provide actionable triage advice.

3.2 Identified Research Gaps

- **Absence of Explainable Multimodal Grounding:** Existing systems rarely combine textual diagnostics with exact visual heatmap verification.
- **High Computational Overhead:** There is a need for quantized, offline-capable vision-language approaches suitable for edge smartphone hardware.
- **Lack of Actionable Triage Systems:** Standard models output probabilities without translating them into useful emergency referral timelines for health workers.

3.3 Datasets

Dataset	Resource	Key Details
ChestX-ray14	NIH CC Box	112,120 chest X-rays with 14 disease labels.
OpenI (OpenAccess)	NLM OpenI	7,470 X-rays paired with clinical radiology reports.
ROCO Medical	GitHub / ROCO	Multimodal radiology dataset for text-image alignment.

4. METHODOLOGY

Stage	Approach
Data-Driven Class Selection	Retain 11 of 14 NIH conditions based on measured per-class support.
Frozen-Backbone Transfer Learning	Use ImageNet-pretrained ViT-B/32 and DenseNet121; fine-tune only final blocks.
Imbalance-Aware Training	Use weighted loss and best-checkpoint-by-F1 selection to counter class imbalance and overfitting.
Post-Hoc Threshold Tuning	Optimize per-class decision thresholds using precision-recall curves without retraining.
Retrieval-Augmented Reporting	Use TF-IDF grounding against real OpenI radiology report text.
Independent Cross-Checks	Use zero-shot CLIP and BLIP v1 as separate secondary signals.

4.1 Methodology Flow

1. Acquire and prepare chest X-ray and report datasets.
2. Select the supported disease conditions from ChestX-ray14.
3. Train/fine-tune ViT-B/32 and DenseNet121 using transfer learning.
4. Handle class imbalance using weighted loss and F1-based checkpoint selection.
5. Tune decision thresholds independently for each condition.
6. Generate visual explanations using attention-based visualization.
7. Retrieve relevant report text using TF-IDF and generate multi-condition reporting.
8. Apply independent CLIP and BLIP v1 cross-checks.
9. Pass the outputs to the rule-based triage component.
10. Present predictions, explanations and reports through the Streamlit application with PDF export.

5. SYSTEM DESIGN

Step	Module	Output
1	Upload	Chest X-ray (PNG / JPG)
2	Classify – ViT-B/32	11-condition prediction
3	Explain + Report	Attention heatmap + TF-IDF retrieval
4	Triage	Rule-based urgency scoring
5	Deliver	Streamlit UI + downloadable PDF report

Two independent branches—vision explainability and text reporting—recombine at the presentation layer, alongside supplementary CLIP and BLIP v1 cross-checks.

5.1 End-to-End Architecture

Chest X-ray → ViT-B/32 Classification → Attention Heatmap + TF-IDF Report Retrieval → Rule-based Triage → Streamlit UI / PDF Report

6. TECHNICAL STACK

Technology	Purpose
Python / PyTorch / torchvision	Model definition, training and inference.
HuggingFace Transformers	CLIP and BLIP v1 cross-checks.
scikit-learn	TF-IDF, precision-recall curves and evaluation metrics.
OpenCV	Heatmap overlays and contour visualization.
pandas	Dataset and report handling.
Streamlit	Interactive web application.
fpdf2	Downloadable PDF reports.
Kaggle / kagglehub	Dataset acquisition.

7. CODE STRUCTURE & IMPLEMENTATION

Pipeline Stage	Key Files	Purpose
Data	dataset.py, count_labels.py	Dataset preparation and class-support analysis.
Model	model.py, train.py	ViT-B/32 transfer learning and imbalance-aware training.
Explainability	gradcam.py	Attention-based explainability, heatmap and contour marking.
Evaluation	evaluate.py, tune_thresholds.py	Model evaluation and per-class threshold tuning.
Reporting	report_index.py, report_generator.py	TF-IDF indexing and multi-condition report generation.
Cross-checks	clip_verify.py, blip_caption.py	Independent CLIP and BLIP v1 verification.
Decision	triage.py	Deterministic urgency scoring.
Delivery	inference.py, report_pdf.py, app/app.py	Inference, Streamlit interface and downloadable PDF report.

8. RESULTS / OUTCOMES

Metric	Result
Conditions Classified	11
Best Macro F1 – tuned	0.304
Real Reports Indexed	7,426

8.1 Backbone Comparison

Model	Default Threshold (0.5) Macro F1	Tuned Threshold Macro F1
DenseNet121	0.254	0.304
ViT-B/32	0.230	0.260

Why ViT scored slightly lower: Vision Transformers lack the built-in locality and translation-invariance biases of CNNs and typically need more training data to match CNN performance. The project used approximately 4,800 training images, making the observed gap an expected outcome rather than an implementation defect.

9. PERFORMANCE EVALUATION & DEEPER ANALYSIS

9.1 Recall Improvement from Per-Class Threshold Tuning

Condition	Default Threshold Recall	Tuned Threshold Recall
Cardiomegaly	0.062	0.75
Mass	0.079	0.45
Pneumonia	0.000	0.33

Trade-off accepted deliberately: for a triage tool, a missed finding is treated as more costly than a false alarm.

9.2 Explainability Validation

Measure	Observed Result
Heatmap standard deviation	0.000 → 0.277
Visualization issue	Gradient-based Grad-CAM collapsed on the ViT because of token homogenization in the mostly-frozen backbone.
Resolution	Attention-based visualization, the standard ViT approach, restored genuine spatial signal.

9.3 Qualitative Validation

On a held-out case with ground truth of Cardiomegaly, Edema and Effusion, the system correctly flagged Cardiomegaly at 99.9% confidence and Effusion after the multi-condition reporting upgrade.

10. PROJECT DESIGN & DEPLOYMENT

Component	Design Choice
Input	Chest X-ray in PNG / JPG format.
Prediction	11-condition ViT-B/32 classifier.
Explainability	Attention-based heatmap and contour visualization.
Reporting	TF-IDF retrieval grounded in OpenI report text.
Triage	Deterministic rule-based urgency scoring.
Application	Streamlit interactive UI.
Output	Prediction, visual explanation, report and downloadable PDF.
Deployment Goal	Offline-oriented rural diagnostic support.

11. EXPECTED PROJECT OUTCOME

- Explainable chest X-ray classification across 11 selected conditions.
- Visual evidence that helps users inspect regions associated with predictions.
- Retrieval-grounded multi-condition diagnostic reporting using real radiology report text.
- Per-class threshold tuning to improve recall for triage-oriented use.
- Rule-based urgency scoring to support prioritization of cases.
- Independent CLIP and BLIP v1 signals for supplementary cross-checking.
- Interactive Streamlit delivery with downloadable PDF reports.
- A practical offline-oriented workflow aimed at reducing rural diagnostic delays.

12. CONCLUSION

The Explainable Vision-Language Diagnostic Assistant combines chest X-ray classification, visual explainability, retrieval-grounded reporting and rule-based triage in a single workflow. The implementation evaluates ViT-B/32 and DenseNet121 using transfer learning, addresses class imbalance through weighted training and threshold tuning, and uses attention-based visualization to obtain meaningful spatial evidence from the ViT model. The resulting system is designed as an offline-oriented decision-support tool for rural healthcare settings, with predictions, explanations, reports and priority guidance delivered through a Streamlit interface.

13. KEY REFERENCES / RESOURCES

- ChestX-ray14 – NIH Chest X-ray dataset.
- OpenI – Indiana University / NLM OpenAccess chest X-ray and report collection.
- ROCO Medical – radiology image-text dataset.
- ResNet / DenseNet – standard deep-learning image-classification architectures used as relevant existing work and baselines.
- CLIP – image-text representation learning used for an independent cross-check.
- BLIP v1 – image captioning / vision-language cross-check.
- TF-IDF – retrieval mechanism for grounding report generation in OpenI report text.

THANK YOU